HDOC – 61th day

Two Sum Using Pointers

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Problem Statement

Write a C program that reads the size and elements of a variable-length integer array, nums[], containing only positive integers, and an integer target. Pass the array to a function named two_sum() using its address. Use pointers to pass the array between the calling function and two_sum(). The function two_sum() must:

- Find two distinct indices, i and j, such that: nums[i] + nums[j] = target.
- Display the two indices separated by a space.
- Return the indices in any order.
- Display -1 -1 if no such pair exists.
- Use a nested-loop approach to solve the problem.
- Assume that at most one valid solution exists.

Algorithm

1. Start the program.

2. Read the size n of the array.

3. Declare a variable-length integer array nums[n].

4. Read n positive integers into the array.

5. Read the value of target.

6. Pass the address of the array, its size, and target to two_sum().

7. In two_sum(), use two nested loops with indices i and j.

8. Start j from i + 1 so that two distinct elements are checked.

9. Access the elements using pointers as *(nums + i) and *(nums + j).

10. Check whether their sum is equal to target.

11. If a pair is found, display i and j and return.

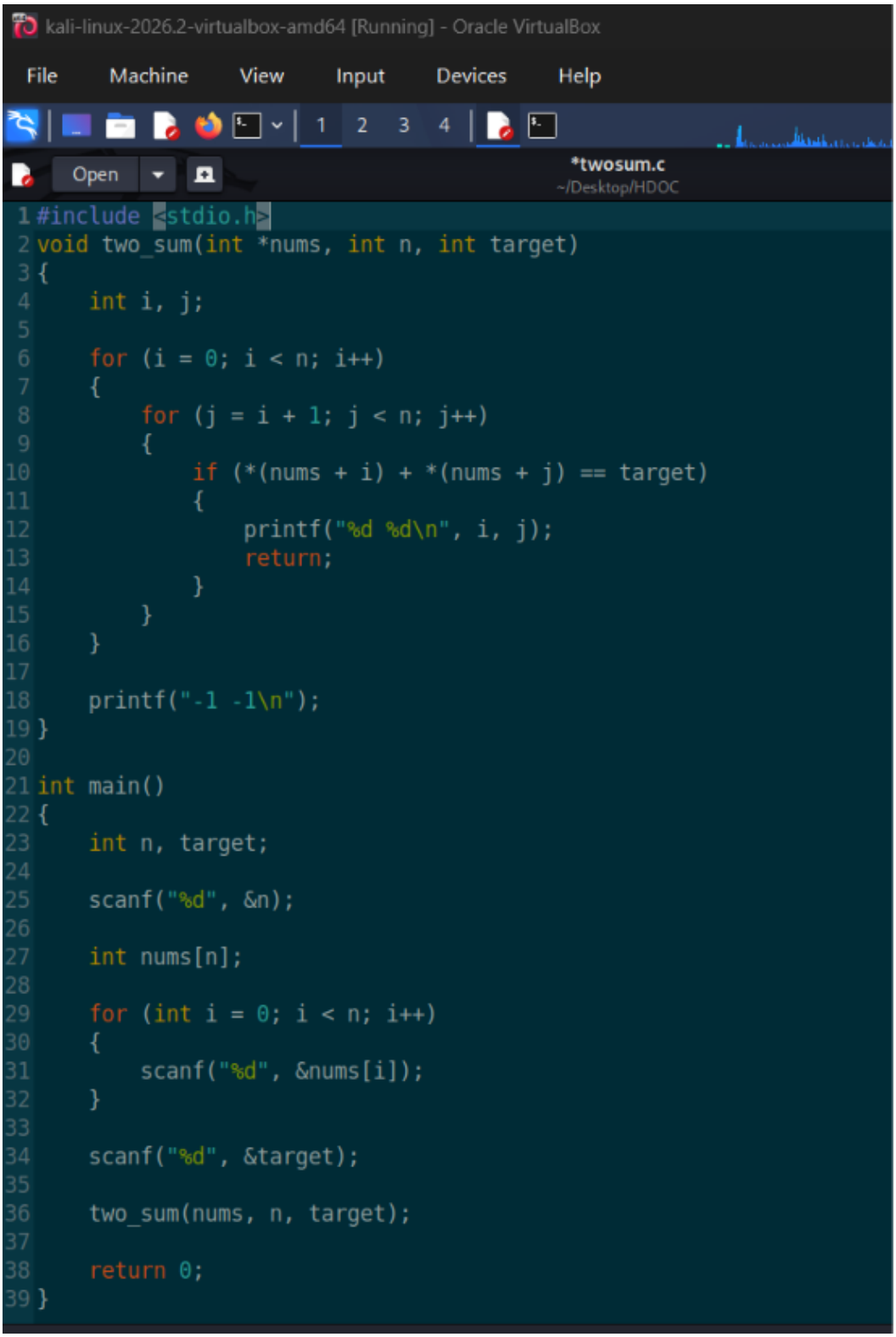
12. If no pair is found after checking all possibilities, display -1 -1.

13. Stop.

Test Case Table

TEST CASE	INPUT	EXPECTED OUTPUT	ACTUAL OUTPUT	RESULT
1	4 2 7 11 15 9	0 1	0 1	Pass
2	3 3 2 4 6	1 2	1 2	Pass
3	4 1 5 3 7 10	2 3	2 3	Pass
4	4 5 10 15 20 25	0 3	0 3	Pass
5	4 1 2 3 4 10	-1 -1	-1 -1	Pass

C Program




```c
#include <stdio.h>
void two_sum(int *nums, int n, int target)
{
    int i, j;

    for (i = 0; i < n; i++)
    {
        for (j = i + 1; j < n; j++)
        {
            if (*(nums + i) + *(nums + j) == target)
            {
                printf("%d %d\n", i, j);
                return;
            }
        }
    }

    printf("-1 -1\n");
}

int main()
{
    int n, target;

    scanf("%d", &n);

    int nums[n];

    for (int i = 0; i < n; i++)
    {
        scanf("%d", &nums[i]);
    }

    scanf("%d", &target);

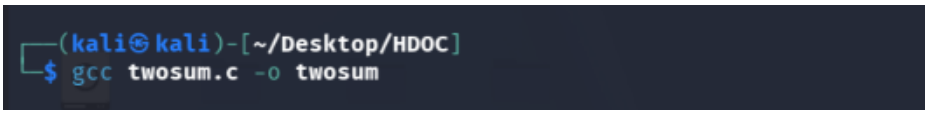
    two_sum(nums, n, target);

    return 0;
}
```

Compilation and Execution

Compile the program using:

gcc twosum.c -o twosum



Execute the program using:

./twosum

